KAVACH (कवच): Updated Project Plan & Progress-vs-Plan Audit

16-Week Agile Milestone Tracking, Slip Analysis & Phase-3 Roadmap

Academic Course: CSE3101 — Agentic AI (3 Credits, LDP: 2-0-2)
Semester & Batch: 7th Semester B.Tech CSE (AI & Data Science) · Academic Year 2026–27
Faculty Mentors: Dr. Soharab Hossain Shaikh (Course Coordinator) & Mr. Pranshu Tiwari
Institution: BML Munjal University, Gurugram
Team Members: Dhruv Jain, Hemant Goyal, Rajat Yadav
Deliverable Type: Mid-Term Evaluation (Phase-2) Deliverable 2 (Rubric C6)

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1. 16-Week Project Execution Schedule: Planned vs. Actual

Week	Milestone / Module Description	Planned Wk	Actual Wk	Status
W01	Threat modeling, PEAS definition & Project Charter	Week 1	Week 1	COMPLETED
W02	Baseline regex PII & secret detection engine	Week 2	Week 2	COMPLETED
W03	Phase-1 Submission & Project Charter Evaluation	Week 3	Week 3	COMPLETED
W04	AST code parser & function chunking prototype	Week 4	Week 4	COMPLETED
W05	Qdrant dense vector store integration (all-MiniLM)	Week 5	Week 5	COMPLETED
W06	10-Stage Finite State Machine (FSM) Orchestrator	Week 6	Week 6	COMPLETED
W07	Bidirectional AST blast-radius analyzer	Week 7	Week 7	COMPLETED
W08	PyPI Slopsquatting & package verification firewall	Week 8	Week 8	COMPLETED
W09	Bounded ReAct self-healing sandbox loop (N <= 3)	Week 9	Week 9	COMPLETED
W10	Model Context Protocol (MCP) & Google A2A protocol	Week 10	Week 10	COMPLETED
W11	Bharat Core Banking Highway & SAGA Governance	Week 11	Week 11	COMPLETED
W12	GitHub PR Guardian Webhook listener integration	Week 12	Roadmap	IN PROGRESS
W13	Multi-tenant PostgreSQL database migration	Week 13	Roadmap	PLANNED
W14	Firecracker / gVisor MicroVM containerization	Week 14	Roadmap	PLANNED
W15	Comprehensive IEEE research publication bundle	Week 15	Roadmap	IN PROGRESS
W16	Final Phase-3 Capstone Presentation & Defense Viva	Week 16	Roadmap	SCHEDULED

2. Deep-Dive Slip Analysis & Architectural Adjustments

In accordance with **Rubric C6 (Progress vs Plan)** and honest senior software engineering principles, the table below documents the engineering slips encountered during Phase-2 implementation, the root causes, and the concrete mitigations executed:

Slip ID	Component / Milestone	Planned Approach	Actual Implementation Reality	Root Cause & Technical Justification	Architectural Mitigation Executed
SLIP-01	**Air-Gapped Local LLM Routing**	Run a live, local `qwen2.5-coder:7b` Ollama daemon on all evaluation workstations.	Graceful Dual-Engine router: Google Gemini 2.5 Flash live cloud inference, switching to simulated zero-network code synthesis when offline.	Running an unquantized 7B/8B model requires > 8 GB dedicated GPU VRAM, which frequently caused out-of-memory crashes on student laptops.	Implemented an air-gapped privacy switch in `app/generation/llm_client.py` that routes to local Ollama when VRAM is available and gracefully activates a deterministic, zero-network code generator otherwise.
SLIP-02	**External Framework Dependencies (CrewAI & ADK)**	Install external `crewai` and `google-adk` PyPI distribution packages via `requirements.txt`.	Engineered clean, in-repo native Python architectural abstractions implementing the exact design patterns (`app/adk/callbacks.py` and `app/crew/flows.py`).	External pip packages brought heavy dependency conflicts, unstable API breaking changes across releases, and bloated container images by > 1.8 GB.	Authored the 8-hook callback lifecycle, agent coordinator, and stateful `@start` / `@listen` flows in native pure Python. This eliminates supply-chain risks and ensures sub-millisecond execution.
SLIP-03	**Windows Concurrent Qdrant File Locking**	Maintain persistent on-disk SQLite/Qdrant vector indexes across concurrent pytest workers.	Configured in-memory `:memory:` vector collection modes during test and verification runs.	Windows OS enforces strict mandatory file locking (`WinError 32`), causing access contention when parallel test suites hit on-disk vector stores simultaneously.	Added environment check (`PYTEST_CURRENT_TEST`); when detected, the system automatically runs Qdrant in ephemeral memory mode, eliminating locking bottlenecks.
SLIP-04	**Scope Expansion: Banking Governance**	Confine platform strictly to software engineering DevOps pull requests.	Expanded scope to include **Bharat Core Banking Highway** mounted at `/bank` with 15 dedicated endpoints.	Realized that the OWASP LLM01 prompt injection and lack of transactional atomicity represent an even more acute risk in financial execution than DevOps.	Architected a complete banking sub-application with SAGA compensations, Maker-Checker 4-eyes controls, and time-lock escrow for transfers > ₹50,000.

3. Team Role Rotation & Phase-Wise Responsibility Matrix

To satisfy the mandatory university guideline from **Page 7 of the Course Handout** ("*Roles must rotate across phases. No student may limit their contribution to a single area*"), tasks are systematically rotated across Phase-1, Phase-2, and Phase-3:

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|                                      PHASE-WISE ROLE ROTATION MATRIX                                  |
+------------------+------------------------------+----------------------+--------------------------------+
| Evaluation Phase | Dhruv Jain                   | Hemant Goyal         | Rajat Yadav                    |
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| **PHASE 1**      | Lead Architecture & FSM      | Pre-Execution Security | Agentic RAG & Qdrant Chunking  |
| (Foundations)    | (State Machine & PEAS)       | (Regex & Secret Engine)| (Dense MiniLM Vector Index)    |
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| **PHASE 2**      | Banking Highway & SAGA Engine | AST Blast Radius Lead  | Frontend Dashboard & Voice STT |
| (Midsem Baseline) | (Time-Lock Escrow & Core Bank)| (Static AST Call Graph)| (Groq Whisper & SSE Telemetry) |
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| **PHASE 3**      | MCP & Interoperability Lead  | Empirical Benchmarking | CI/CD Webhooks & Merkle Ledger |
| (Final Capstone) | (Cursor/Claude JSON-RPC Bus) | (V2 Held-Out & Attacks)| (GitHub Actions Pre-Merge Gate)|
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4. Phase-3 Final Sprint Roadmap (Weeks 12 to 16)

The remaining sprints leading to the final capstone defense focus on cloud production readiness: 1. **Sprint 12 (GitHub PR Guardian Bot):** Complete webhook listener (`/webhook/github`) that posts automated AST blast-radius and slopsquatting annotations directly into GitHub Pull Request threads. 2. **Sprint 13 (Production PostgreSQL Migration):** Replace local SQLite session persistence with multi-tenant PostgreSQL managed via SQLAlchemy and Alembic migrations. 3. **Sprint 14 (Firecracker MicroVM Isolation):** Graduate the ephemeral subprocess test runner into hardware-isolated microVM sandboxing for untrusted third-party code execution. 4. **Sprint 15 (IEEE Conference Manuscript Finalization):** Finalize LaTeX publication draft titled "*Deterministic Pre-Execution Governance and Supply-Chain Firewalls for Autonomous AI Coding Agents*". 5. **Sprint 16 (Final Capstone Defense & Viva):** Execute the 6-minute live presentation and demonstrate 100% test pass rate to the examination panel.